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Steering control apparatus

Abstract

A steering control apparatus comprises an environment sensor for detecting a condition around a vehicle; and a controller for controlling turning of a front wheel and turning of a rear wheel, wherein the controller is configured to: calculate a total steering angle which is a steering angle required for causing the vehicle to travel along a target path, based on at least a detection result of the environment sensor; turn the front wheel based on a front wheel target steering angle; and turn the rear wheel based on a rear wheel target steering angle that is a subtraction value obtained by subtracting the front wheel target steering angle from the total steering angle.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-19083 filed on Feb. 10, 2023, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

(2) This specification discloses a steering control apparatus for controlling steering of a vehicle.

BACKGROUND

(3) There has been known a technology for automatically steering a rear wheel in order to improve traveling performance of a vehicle. For example, Patent Document 1 discloses a technology in which a rear wheel is automatically steered in order to compensate for a self-aligning torque generated in a front wheel, whereby the front wheel can maintain its original steering angle.

(4) However, many of such prior arts are based on the assumption that the driver steers the front wheel. Therefore, the technology of Patent Document 1 cannot be applied to an automatic steering technology in which the vehicle automatically performs all steering of the vehicle.

(5) In other words, recently, there has been known an automatic steering technology in which a

vehicle automatically performs all steering of a vehicle. For example, in the lane keep assist function, the vehicle automatically controls a steering angle of a wheel so that the vehicle runs along a lane. In order to improve the traveling performance of a vehicle, it is also required to appropriately steer a rear wheel in such an automatic steering technology. However, in the conventional automatic steering technology, steering of the rear wheel is not sufficiently studied. As a result, in the conventional automatic steering technology, the traveling performance of the vehicle cannot be sufficiently improved, or there is a possibility that the driver feels uncomfortable with respect to steering.

(6) Accordingly, this specification discloses a steering control apparatus capable of further improving the traveling performance of a vehicle while reducing the feeling of discomfort of an occupant during automatic steering.

CITATION LIST

(7) PATENT DOCUMENT 1: JP. 6706634 B

SUMMARY

(8) A steering control apparatus disclosed herein comprises an environment sensor for detecting a condition around a vehicle; and a controller for controlling turning of a front wheel and turning of a rear wheel, wherein the controller is configured to: calculate a total steering angle which is a steering angle required for causing the vehicle to travel along a target path, based on at least a detection result of the environment sensor; turn the front wheel based on a front wheel target steering angle; and turn the rear wheel based on a rear wheel target steering angle that is a subtraction value obtained by subtracting the front wheel target steering angle from the total steering angle.

(9) With such a configuration, since the steering angle of the vehicle is distributed to the front wheel and the rear wheel as in the case of the manual steering, it is possible to improve the traveling performance of the vehicle while reducing the feeling of discomfort of the occupant.

(10) The controller may be configured to: calculate the target path and a current actual path based on the detection result of the environment sensor; calculate an FF steering angle based on the target path; calculate an FB steering angle based on a difference between the target path and the actual path; and calculate a sum of the FF steering angle and the FB steering angle as the total steering angle, the front wheel target steering angle is the FF steering angle, and the rear wheel target steering angle is the FB steering angle.

(11) With such a configuration, the feeling of discomfort of the occupant can be reduced with a simple configuration.

(12) The controller may be configured to: calculate the target path and a current actual path based on the detection result of the environment sensor; calculate an FF steering angle based on the target path; calculate an FB steering angle based on a difference between the target path and the actual path; and calculate a sum of the FF steering angle and the FB steering angle as the total steering angle, the rear wheel target steering angle is a value obtained by multiplying the FB steering angle by a distribution coefficient of greater than 0 and less than 1, a value obtained by multiplying the total steering angle by the distribution coefficient, or a value obtained by adding a portion of the FF steering angle to a value obtained by multiplying the FB steering angle by the distribution coefficient, and the front wheel target steering angle is a subtraction value obtained by subtracting the rear wheel target steering angle from the total steering angle.

(13) With such a configuration, a steering feeling similar to that of the manual steering can be given to the occupant. As a result, the uncomfortable feeling of the occupant can be further reduced.

(14) The distribution coefficient may be a variable value that decreases as at least one of a lateral acceleration, a vehicle speed, and a curvature of the target path increases.

(15) With this configuration, even when the lateral acceleration is large, the same steering feeling as that of the manual steering can be given to the occupant. As a result, the uncomfortable feeling of the occupant can be further reduced.

(16) The controller may be configured to: calculate a front wheel FB torque based on a difference between the front wheel target steering angle and a detected steering angle of the front wheel; calculate a front wheel FF torque for compensating for a self-aligning torque of the front wheel and a self-aligning torque of the rear wheel based on the total steering angle; and calculate a front wheel torque command for turning the front wheel based on a sum of the front wheel FF torque and the front wheel FB torque.

(17) With this configuration, the front wheel can be more appropriately steered.

(18) According to the steering control apparatus disclosed in this specification, since the total steering angle is distributed to the front wheel and the rear wheel to turn the steering wheel, it is possible to improve the traveling performance of the vehicle during automatic steering while reducing the feeling of discomfort of the occupant.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Embodiment(s) of the present disclosure will be described based on the following figures, wherein:

(2) FIG. 1 is a block diagram showing a configuration of a steering control apparatus;

(3) FIG. 2 is a diagram showing movement paths due to steering of front wheels and rear wheels;

(4) FIG. 3 is a functional block diagram of a controller;

(5) FIG. 4 is a control block diagram showing a flow of calculation of a front wheel torque command and a rear wheel torque command; and

(6) FIG. 5 is a control block diagram showing another example of the flow of calculation of the front wheel torque command and the rear wheel torque command.

DESCRIPTION OF EMBODIMENT

(7) Hereinafter, the configuration of the steering control apparatus **10** will be described with reference to the drawings. FIG. 1 is a block diagram showing a configuration of a steering control apparatus **10**. The steering control apparatus **10** is applied to an automatic driving vehicle in which the vehicle automatically controls steering and acceleration/deceleration, or an advanced driving assist vehicle in which at least the vehicle temporarily controls steering. Advanced driving assistance includes lane keep assist that automatically travels the vehicle along a lane.

(8) The steering control apparatus **10** includes a plurality of sensors **12**, **14**, and **16**, actuators **18** and **20** for driving wheels, and a controller **26** for controlling the actuators **18** and **20**. The environment sensor **12** detects a state around the vehicle. More specifically, the environment sensor **12** detects at least a lane and a surrounding obstacle (e.g., another vehicle or a walker). The environment sensor **12** includes, for example, at least one of a camera, a LiDAR, a millimeter wave radar, and an ultrasonic sensor.

(9) The vehicle sensor **14** detects a state of the vehicle. More specifically, the vehicle sensor **14** detects at least a position, a vehicle speed, an acceleration, and six-component force at the center of gravity. Note that the six-component force at the center of gravity include a longitudinal force, a lateral force, a vertical force, a roll moment, a pitch moment, and a yaw moment. The vehicle sensor **14** includes, for example, a position sensor (e.g., GPS) and an inertial sensor (e.g., a triaxial acceleration sensor, a triaxial gyroscope, a triaxial geomagnetic sensor, etc.). The steering angle sensor **16** detects a steering angle of the front wheel **22**.

(10) The front wheel actuator **18** controls a steering angle of the front wheel **22** of the vehicle in accordance with a control command from the controller **26**. The front wheel actuator **18** may be provided in the steering section of the vehicle as an assist motor which is a part of the electric power steering system. Further, the front wheel actuator **18** may be provided as a steer-by-wire mechanically independent of the steering section of the vehicle.

(11) The rear wheel actuator **20** controls a steering angle of the rear wheel **24** of the vehicle in accordance with a control command from the controller **26**. The rear wheel actuator **20** may be provided in the steering section of the vehicle as an assist motor, or may be provided mechanically independent of the steering section of the vehicle as a steer-by-wire.

(12) The controller **26** outputs a control command for controlling steering of the front wheel **22** and the rear wheel **24** based on the detection results of the plurality of sensors **12**, **14**, and **16**. The controller **26** is physically a computer having a processor **28** and a memory **30**. Although the controller **26** is illustrated in FIG. **1** as a single computer, the controller **26** may comprise a plurality of physically separate computers. For example, the controller **26** may include an electric power steering ECU and a dynamic rear steering (hereinafter referred to as “DRS”). The electric power steering ECU is a computer that controls steering of the front wheel **22**. The DRS is a computer that controls steering of the rear wheel **24**.

(13) Next, steering control by the steering control apparatus **10** will be described. The steering control described below is executed for automatic driving or advanced driving support for temporarily automatically controlling steering.

(14) In the steering control, the controller **26** controls the steering angles of the front wheel **22** and the rear wheel **24** so that the vehicle passes through the center of the lane **50** on which the vehicle is currently traveling. Therefore, the center line of the lane **50** is the target path of the vehicle. The controller **26** controls the steering angle of the front wheel **22** so that the theoretical travel path of the vehicle has the same shape as the target path. Here, even if the theoretical moving path has the same shape as the target path, the actual moving path may deviate from the target path due to disturbance (for example, wind, obstacles on a road surface, or the like), intervention by a driver, or the like. Accordingly, the controller **26** compensates for the shift between the theoretical moving path and the actual path by turning the rear wheel **24**.

(15) FIG. **2** is a diagram showing a movement path due to steering of each of the front wheel **22** and the rear wheel **24**. In FIG. **2**, the solid line indicates the actual path **52** of the vehicle. The two-dot chain line indicates a theoretical movement path obtained by turning the front wheel **22**. Hereinafter, this moving path is referred to as a “front wheel path **54**”. The broken line indicates a movement path obtained by turning the rear wheel **24**. Hereinafter, this movement path is referred to as a “rear wheel path **56**”.

(16) As shown in FIG. **2**, when the lane **50** is linear, the front wheel path **54** is also linear. In this case, the steering angle of the front wheel **22** is “0”. However, a difference may occur between the actual path **52** and the target path due to disturbance or operator intervention. At this time, the controller **26** turns the rear wheel **24** so as to resolve the difference.

(17) When the lane **50** is curved, the front wheel path **54** is also curved. In this case, the steering angle of the front wheel **22** is an angle corresponding to the curvature of the lane **50**. Also in this case, a difference may occur between the actual path **52** and the target path. Then, the controller **26** turns the rear wheel **24** so as to resolve the difference.

(18) By turning the front wheel **22** in accordance with the target path and absorbing the difference from the target path by turning the rear wheel **24**, the comfort of the occupant can be improved. For example, when the turning amount of the front wheel **22** is large during the straight traveling, the occupant tends to feel uncomfortable. According to the above-described control, since the turning amount of the front wheel **22** is reduced during the straight traveling, the uncomfortable feeling of the occupant is reduced, and the comfort of the occupant is improved. Further, even when the vehicle runs on a curve, if the turning amount of the front wheel **22** largely deviates from the curvature of the lane **50**, the occupant tends to feel uncomfortable. According to the above-described control, since the turning amount of the front wheel **22** during the curve traveling is substantially the same as the turning amount in the case of the manual steering, the feeling of discomfort of the occupant is reduced. As a result, the comfort of the occupant is improved.

(19) Next, the steering control process will be described with reference to FIG. **3**. FIG. **3** is a

functional block diagram of the controller **26**. The target yaw rate calculation unit **32** calculates a target yaw rate YR based on the detection results of the environment sensor **12** and the vehicle sensor **14**. The target yaw rate YR is a parameter indicating a target path. The target yaw rate YR is calculated from the vehicle speed Vv and the curvature of the lane **50**. The curvature of the lane **50** is obtained by analyzing data detected by the environment sensor **12** (e.g., an image captured by a camera).

(20) The FF steering angle calculation unit **34** calculates a feedforward steering angle θ_{ff} (hereinafter referred to as “FF steering angle θ_{ff} ”) based on the target yaw rate YR. The FF steering angle θ_{ff} is a steering angle required for moving the vehicle at the target yaw rate YR. In other words, the FF steering angle θ_{ff} is a steering angle at which the theoretical travel path of the vehicle has the same shape as the target path. In this example, the FF steering angle θ_{ff} is the front wheel target steering angle θ_{F*} .

(21) The FB steering angle calculation unit **36** calculates a feedback steering angle θ_{fb} (hereinafter referred to as “FB steering angle θ_{fb} ”). The FB steering angle θ_{fb} is a steering angle for eliminating the difference between the target path and the actual path **52**. The FB steering angle θ_{fb} is calculated based on, for example, a lateral position, a lateral speed, a yaw angle, a yaw rate, and a steering angle. In this example, the FB steering angle θ_{fb} is the rear wheel target steering angle θ_{R*} .

(22) The controller **26** adds the FF steering angle θ_{ff} and the FB steering angle θ_{fb} to calculate the total steering angle θ . The total steering angle θ is a steering angle required for the vehicle to travel along the target path. The FF steering angle θ_{ff} and the total steering angle θ are input to the electric power steering ECU **38**. The electric power steering ECU **38** calculates a front wheel torque command TF for turning the front wheel **22** based on the FF steering angle θ_{ff} and the total steering angle θ . The front wheel actuator **18** turns the front wheel in accordance with the front wheel torque command TF. The flow of calculation of the front wheel torque command TF will be described later.

(23) Further, the controller **26** calculates the rear wheel break angle θ_{8R} based on the FB steering angle θ_{fb} and the steering gear ratio NS. The DRS **40** calculates the rear wheel torque command TR based on the rear wheel break angle θ_{8R} . The rear wheel actuator **20** turns the rear wheel **24** in accordance with the rear wheel torque command TR.

(24) FIG. **4** is a control block diagram showing a flow of calculation of the front wheel torque command TF and the rear wheel torque command TR. As shown in FIG. **4**, the front wheel torque command TF is a value obtained by adding the front wheel feedback torque TFfb (hereinafter referred to as “front wheel FB torque TFfb”), the front wheel feedforward torque TFff (hereinafter referred to as “front wheel FF torque TFff”), and the front wheel steering torque TFsf.

(25) The front wheel FB torque TFfb is a steering torque for setting the front wheel **22** to the front wheel target steering angle θ_{F*} . In the example of FIG. **4**, the front wheel target steering angle θ_{F*} is the FF steering angle θ_{ff} . The controller **26** calculates a steering angle difference $\Delta\theta$, which is a subtraction value obtained by subtracting the front wheel detected steering angle θ_{Fd} from the FF steering angle θ_{ff} . Further, the controller **26** calculates the front wheel FB torque TFfb by multiplying the steering angle difference $\Delta\theta$ by a predetermined gain G. Here, the front wheel FB torque TFfb is calculated by proportional calculation. However, the front wheel FB torque TFfb may be calculated by a proportional integral operation or a proportional integral differential operation.

(26) The front wheel FF torque TFff is a steering torque for compensating for a self-aligning torque generated when the front wheel **22** and the rear wheel **24** are steered. The front wheel FF torque TFff is calculated by the following equation (1). In the equation (1), Kh is a stability factor, l is a wheel base, Gff is a vehicle speed gain, and T is a torque conversion coefficient.

$$(27) \quad TF_{ff} = \frac{V_v^2 \times \times T \times G_{ff}}{(1 + K_h \times V_v^2) \times N_s \times l} \quad (1)$$

(28) The front wheel steering torque TF_{sf} is calculated according to the driver torque and the steering angle, and is an assist torque for assisting the driver. The driver torque is torque applied as a steering input by the driver. Since the front wheel steering torque TF_{sf} can be calculated by a conventional technique, a detailed description thereof is omitted here.

(29) The controller **26** adds the front wheel FB torque TF_{fb} , the front wheel FF torque TF_{ff} , and the front wheel steering torque TF_{sf} , and outputs a front wheel torque command TF . The front wheel actuator **18** turns the front wheel in response to the front wheel torque command TF .

(30) The rear wheel torque command TR is calculated based on the rear wheel target steering angle θ_{R^*} . In this example, the rear wheel target steering angle θ_{R^*} is the FB steering angle θ_{fb} . Here, the FB steering angle θ_{fb} is an indication angle with respect to the front wheel **22**. Since the rear wheel **24** is controlled in the opposite phase to the front wheel **22**, the rear wheel break angle θ_{R} for realizing the FB steering angle θ_{fb} is a value obtained by multiplying the FB steering angle θ_{fb} by $(-1/N_s)$. The DRS **40** multiplies the rear wheel break angle SR by a predetermined coefficient to calculate a rear wheel torque command TR . The rear wheel actuator **20** steers the rear wheel **24** in accordance with the rear wheel torque command TR .

(31) As apparent from the above description, the controller **26** performs feedback control so that the front wheel **22** becomes the FF steering angle θ_{ff} . The self-aligning torque is also compensated by the front wheel **22**. The controller **26** compensates for the difference between the target path and the actual path **52** by steering the rear wheel **24**. As a result, even during the automatic driving or the execution of the advanced driving support, the occupant can feel the same feeling as when the vehicle is manually steered. As a result, the comfort of the occupant is improved. Since the difference between the paths is compensated by steering of the rear wheel **24**, the vehicle runs along the target path.

(32) Incidentally, when the vehicle runs on a curve, if the correction of the path difference (hereinafter referred to as “corrected steering”) is performed only by the steering of the rear wheel **24**, the occupant may feel uncomfortable. Therefore, depending on the vehicle type and the traveling condition, a part of the FB steering angle θ_{fb} for the corrected steering may be placed on the front wheel **22**.

(33) FIG. 5 is a control block diagram in this case. As shown in FIG. 5, in this case, the rear wheel target steering angle θ_{R^*} is a value obtained by multiplying the FB steering angle θ_{fb} by a predetermined distribution coefficient K . That is, $\theta_{R^*} = \theta_{fb} \times K$. The distribution coefficient K satisfies $0 < K < 1$. The front wheel target steering angle θ_{F^*} is a value obtained by adding a value obtained by multiplying the FB steering angle θ_{fb} by $(1-K)$ and the FF steering angle θ_{ff} . That is, $\theta_{F^*} = \theta_{ff} + \theta_{fb}(1-K)$. The example of FIG. 5 differs from the example of FIG. 4 in the values of the front wheel target steering angle θ_{F^*} and the rear wheel target steering angle θ_{R^*} , but the flow of other calculations is the same as the example of FIG. 4.

(34) By distributing a part of the FB steering angle θ_{fb} to the front wheel **22** in this manner, the uncomfortable feeling of the occupant may be further reduced. The distribution coefficient K may be an invariant constant or a variable. For example, the distribution coefficient K may be a variable value that decreases as at least one of the lateral acceleration, the vehicle speed V_v , and the curvature of the target path increases. The reason for this is as follows. Normally, when steering manually, when the driver feels a lateral acceleration, the driver operates the steering wheel to turn the front wheel in order to cancel the influence of the lateral acceleration. Also in automatic steering, it is required to give the driver a steering feeling similar to that of the manual steering. Therefore, during execution of the automatic operation, when the lateral acceleration is large, the distribution coefficient K is reduced, and a part of the corrected steering is placed on the front wheel **22**. Thus, the feeling of discomfort of the occupant can be reduced. The greater the vehicle

speed V_v and the curvature of the target path, the greater the lateral acceleration. Therefore, the distribution coefficient K may be changed according to the vehicle speed V_v and the curvature.

(35) Further, as another form, not only the FB steering angle θ_{fb} but also the FF steering angle θ_{ff} may be distributed to the front wheel **22** and the rear wheel **24**. For example, the front wheel target steering angle θ_{F^*} may be $\theta_{ff}(1-P)+\theta_{fb}(1-K)$. Further, the rear wheel target steering angle θ_{R^*} may be $\theta_{ff}\times P+\theta_{fb}\times K$. Note that P is a coefficient that satisfies $0<P<1$, and is independent of the distribution coefficient K .

(36) Further, depending on the vehicle, the FF steering angle θ_{ff} and the FB steering angle θ_{fb} may not be calculated individually. In other words, the total steering angle θ may be directly calculated from the detection values of the environment sensor **12** and the vehicle sensor **14** for the purpose of reducing the calculation amount. In this case, the total steering angle θ may be distributed to the front wheel **22** and the rear wheel **24** at a predetermined ratio. That is, the front wheel target steering angle θ_{F^*} may be set to $\theta(1-K)$, and the rear wheel target steering angle θ_{R^*} may be set to $\theta\times K$.

(37) With such a configuration, even when the FF steering angle θ_{ff} and the FB steering angle θ_{fb} cannot be calculated individually, steering can be appropriately performed. However, in this case, the distribution ratio of the FF steering angle θ_{ff} and the distribution ratio of the FB steering angle θ_{fb} become the same. In this case, when the distribution coefficient K is set to a constant value, there is a possibility that the vehicle body slip angle at the time of the curve traveling greatly changes from the vehicle body slip angle at the time of the manual steering. When the vehicle body slip angle greatly changes from the vehicle body slip angle at the time of manual steering, the occupant feels uncomfortable. In order to reduce such an uncomfortable feeling, the distribution coefficient K may be a variable value that varies according to at least one of the lateral acceleration, the vehicle speed V_v , and the curvature of the target path. That is, the distribution coefficient K may be decreased as at least one of the lateral acceleration, the vehicle speed V_v , and the curvature of the target path increases. Of course, the distribution coefficient K may be a fixed value.

(38) Any of the configurations described above is an example. As long as the steering control apparatus **10** has the configuration described in claim **1**, other configurations may be appropriately changed. For example, in the above description, the front wheel torque command TF includes the front wheel FB torque TF_{fb} , the front wheel FF torque TF_{ff} , and the front wheel steering torque TF_{sf} . However, if the front wheel torque command TF includes at least the front wheel FF torque TF_{ff} for achieving the front wheel target steering angle θ_{F^*} , at least one of the front wheel FF torque TF_{ff} and the front wheel steering torque TF_{sf} may be omitted. The front wheel torque command TF may further include another compensation torque. The rear wheel torque command TR may also include another compensation torque.

REFERENCE SIGNS LIST

(39) **10** steering control apparatus, **12** environment sensor, **14** vehicle sensor, **16** steering angle sensor, **18** front wheel actuator, **20** rear wheel actuator, **22** front wheel, **24** rear wheel, **26** controller, **28** processor, **30** memory, **32** target yaw rate calculation unit, **34** FF steering angle calculation unit, **36** FB steering angle calculation unit, **38** electric power steering ECU, **40** DRS, **50** lane, **52** actual path, **54** front wheel path, **56** rear wheel path, FB front wheel, FF front wheel, TF front wheel torque command, TF_{fb} front wheel feedback torque, TF_{ff} front wheel feedforward torque, TF_{sf} front wheel steering torque, TR rear wheel torque command, $\Delta\theta$ steering angle difference, δR rear wheel break angle, θ total steering angle, θ_{F^*} front wheel target steering angle, θ_{Fd} front wheel detected steering angle, θ_{R^*} rear wheel target steering angle, θ_{fb} FB steering angle, θ_{ff} FF steering angle, V_v vehicle speed, YR target yaw rate.

Claims

1. A steering control apparatus comprising: an environment sensor for detecting a condition around a vehicle; and a controller for controlling turning of a front wheel and turning of a rear wheel, wherein the controller is configured to: calculate a total steering angle which is a steering angle required for causing the vehicle to travel along a target path, based on at least a detection result of the environment sensor; turn the front wheel based on a front wheel target steering angle; and turn the rear wheel based on a rear wheel target steering angle that is a subtraction value obtained by subtracting the front wheel target steering angle from the total steering angle.
 2. The steering control apparatus according to claim 1, wherein the controller is configured to: calculate the target path and a current actual path based on the detection result of the environment sensor; calculate an FF steering angle based on the target path; calculate an FB steering angle based on a difference between the target path and the actual path; and calculate a sum of the FF steering angle and the FB steering angle as the total steering angle, the front wheel target steering angle is the FF steering angle, and the rear wheel target steering angle is the FB steering angle.
 3. The steering control apparatus according to claim 1, wherein the controller is configured to: calculate the target path and a current actual path based on the detection result of the environment sensor; calculate an FF steering angle based on the target path; calculate an FB steering angle based on a difference between the target path and the actual path; and calculate a sum of the FF steering angle and the FB steering angle as the total steering angle, the rear wheel target steering angle is a value obtained by multiplying the FB steering angle by a distribution coefficient of greater than 0 and less than 1, a value obtained by multiplying the total steering angle by the distribution coefficient, or a value obtained by adding a portion of the FF steering angle to a value obtained by multiplying the FB steering angle by the distribution coefficient, and the front wheel target steering angle is a subtraction value obtained by subtracting the rear wheel target steering angle from the total steering angle.
 4. The steering control apparatus according to claim 3, wherein the distribution coefficient is a variable value that decreases as at least one of a lateral acceleration, a vehicle speed, and a curvature of the target path increases.
 5. The steering control apparatus according to claim 1, wherein the controller is configured to: calculate a front wheel FB torque based on a difference between the front wheel target steering angle and a detected steering angle of the front wheel; calculate a front wheel FF torque for compensating for a self-aligning torque of the front wheel and a self-aligning torque of the rear wheel based on the total steering angle; and calculate a front wheel torque command for turning the front wheel based on a sum of the front wheel FF torque and the front wheel FB torque.
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